

WNC Explore Map

A static, GitHub Pages-ready progressive web app for discovering waterfalls, overlooks, easy family stops, campgrounds, and towns in Western North Carolina.

Included files

- `index.html` — the complete Leaflet map UI, styles, app logic, filters, responsive layout, and built-in points-of-interest dataset.
- `manifest.webmanifest` — install metadata for the PWA.
- `sw.js` — service worker that caches the application shell, Leaflet assets, and map tiles that have been viewed.
- `app-icon.svg` — app icon used by the manifest and Apple touch-icon declaration.

Publish with GitHub Pages

1. Create a new GitHub repository, or open the repository where you want to host the app.
2. Put the four included files in the repository root. Their names must remain exactly as provided.
3. Commit and push the files to GitHub.
4. In the repository, open **Settings** → **Pages**.
5. Under **Build and deployment**, choose **Deploy from a branch**.
6. Select the `main` branch and `/` (root) folder, then save.
7. Wait for GitHub Pages to finish its deployment. Open the published site using the URL shown in the Pages settings.

The app requires HTTPS for full PWA and location functionality. GitHub Pages provides HTTPS automatically.

Install on iPhone

1. Open the live GitHub Pages URL in Safari, not an in-app browser.
2. Tap the Share button.
3. Choose **Add to Home Screen**.
4. Give it a name and tap **Add**.

The installed app opens without browser chrome. iOS supports service-worker caching, but iOS may periodically clear website storage; opening the app online before a trip refreshes its code and previously viewed map tiles.

How offline use works

The application interface and Leaflet library are cached after the first load. Map tiles from OpenStreetMap are cached as you view them. This means the exact areas you pan around while online are usually available later without signal. It is not a full offline map download system.

Before heading into Pisgah, open the map on Wi-Fi, zoom in on every area you plan to visit, and pan around the intended route. Do not bulk-download or programmatically prefetch OpenStreetMap tiles; that violates normal public tile-use expectations.

Editing points of interest

All map data is in one constant near the start of the script in `index.html`:

```
const POIS = [ ... ];
```

Each item needs these fields:

```
{
  n: 'Place name',
  c: 'waterfall', // waterfall, view, family, camp, or town
  e: '☒',
  lat: 35.0000,
  lng: -82.0000,
  r: 'Region',
  kid: 'Short family-facing description',
  dog: true,
  note: 'Practical detail shown in the popup.'
}
```

After changing `sw.js`, increment the cache version at the top, for example from `wnc-explore-v1` to `wnc-explore-v2`. That prompts installed copies to retrieve the new application shell.

Data and safety notes

The map is a trip-ideas tool, not a trail-condition or navigation authority. Each popup has a **Check status** link because access, seasonal operations, pet rules, fees, parking, road closures, weather, and trail conditions can change. Confirm those details with the managing agency before leaving camp.

Directions open Google Maps because it is commonly installed on iOS. The visual map itself uses the open-source Leaflet library and OpenStreetMap tiles. If you prefer Apple Maps, replace the `directions` URL in the `popup()` function with an Apple Maps link format.

Technology

- Leaflet 1.9.4, MIT-licensed open-source mapping library
- OpenStreetMap standard raster tiles
- Plain HTML, CSS, and JavaScript — no build step, framework, API key, analytics, or database
- Responsive bottom sheet optimized for iPhone safe areas
- Search, category filters, accessible keyboard controls, current-location button, marker popups, driving links, and reduced-motion support